



Alice Apple Tree

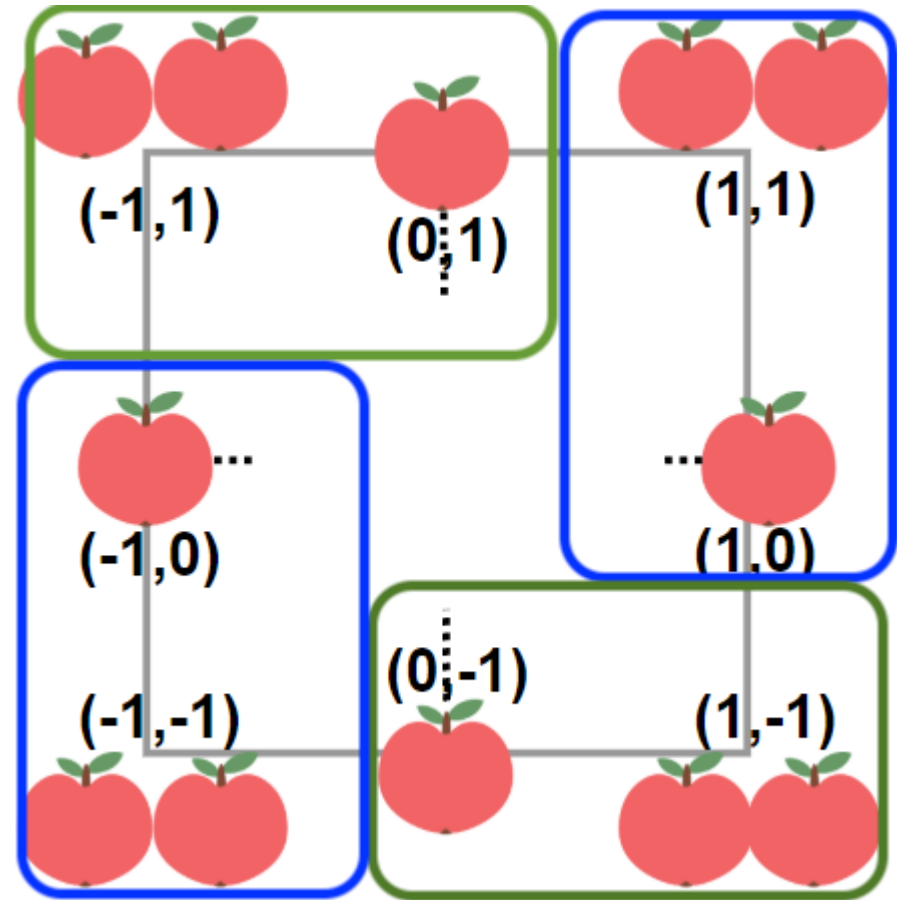
Assume the vertex is (r,r) , $(r,-r)$, $(-r,r)$ and $(-r,-r)$.
Split the apples into 4 rectangles,
each rectangle is $r * (r + 1)$.

Put two blue matrix together
and merge apples tiles at the same position,
each tile become $r + r + 1$.
The total apples in blue is,
 $r * (r + 1) * (r + r + 1)$

Put two green matrix together
and merge apples tiles at the same position,
each tile become $r + r + 1$.
The total apples in green is,
 $r * (r + 1) * (r + r + 1)$

The total apples is:
 $r * r * r * 4 + r * r * 6 + r * 2$.

Then we apply binary search.



									m	n 12*m*m	N Sigma(n)
212 101 212									1	12	12
43234 32123 21012 32123 43234									2	48	60
6543456 5432345 4321234 3210123 4321234 5432345 6543456									3	108	168
876545678 765434567 654323456 543212345 432101234 5432345 6543456 7654567 8765678									4	192	360

```
1. import java.util.*;
2. public class AliceAppleTree {
3.     public static void main(String[] args) {
4.         Scanner sc=new Scanner(System.in);
5.         int apple=sc.nextInt();
6.         int cnt=0,sum=0;
7.         while(sum<apple){
8.             cnt++;
9.             sum+=(12*cnt*cnt);
10.
11.         }
12.         System.out.println((8*(cnt)));
13.     }
14. }
```

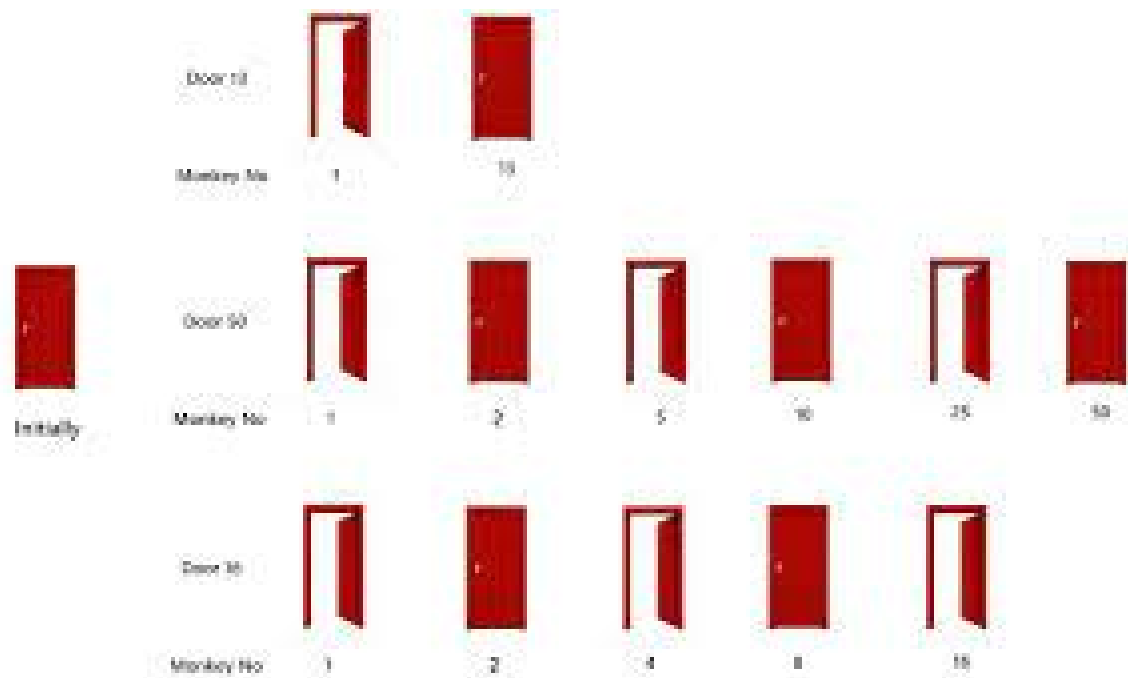
Toggle The Switch

Problem statement

There are 100 closed doors. A cage holding 100 monkeys is placed nearby. Every monkey that visits a door either opens a closed door or closes an open door. The first monkey that is let out of the cage visits and opens all the hundred doors. The second monkey that is released visits doors of the order 2, 4, 6, 8, 10.... . The third monkey released visits doors 3, 6, 9, 12, 15....., and so on. After all the monkeys from the cage are released and have opened or closed at least one door, how many doors are left open?

Approach

Let us understand how to solve this problem by observing the activity of doors 13, 50, and 16. 13 is a prime number, 50 is a composite number, and 16 is a perfect square.



```

1. import java.util.*;
2. public class Toggle {
3.     public static void main(String[] args) {
4.         Scanner sc = new Scanner(System.in);
5.         int n = sc.nextInt();
6.         boolean b[] = new boolean[n+1];
7.         int i,j,c=0,o=0;
8.
9.         for(i=1;i<=n;i++)
10.        {
11.            for(j=i;j*i<=n;j++)
12.            {
13.                if(b[j]==false)
14.                {
15.                    b[j]=true;
16.                }
17.                else{
18.                    b[j]=false;
19.                }
20.            }
21.        }

```

```

1.         for(i=1;i<=n;i++)
2.         {
3.             if(b[i]==true)
4.             {
5.                 c++;
6.             }
7.             else{
8.                 o++;
9.             }
10.        }
11.        System.out.println("No Of Doors
12.        open"+c);
13.        System.out.println("No Of Doors
14.        closed"+o);

```



THANK YOU